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Pool Cleaning Assembly Kit

Abstract

A pool cleaning assembly kit includes a utility connector, a handle base and one or more utility tool. The utility connector includes a connecting dock and two end connectors integrally provided at two ends of the connecting dock respectively. The handle base includes a handle frame detachably connected to the utility connector and configured to be detachably connected with a handle. The one or more utility tool is configured to be detachably connected to at least one of the end connectors so as to integrally assemble with the utility connector to form a cleaning tool.

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Background/Summary

CROSS REFERENCE OF RELATED APPLICATION [0001] This application is a Continuation-In-Part application that claims the benefit of priority under 35 U.S.C. § 120 to a non-provisional application, application Ser. No. 29/899,532, filed Aug. 9, 2023, and a non-provisional application, application Ser. No. 29/899,619, filed Aug. 10, 2023, which are incorporated herewith by references in their entireties.

BACKGROUND OF THE PRESENT INVENTION

Field of Invention

[0002] The present invention relates to a utility assembly, and more particularly to a pool utility assembly kit, which is configured to be assembled to form a pool brush with selectively adapted to be connected with various types of handle bar with different lengths and to connected with various types of brush heads to form a pool brush head with different shape and/or length.

Description of Related Arts

[0003] Pool floor scrub brush (as shown in FIG. 1), pool wall, step and tile brush (as shown in FIG. 2), pool corner brush (as shown in FIG. 3) are utility tools generally required to clean a swimming pool. The user not only needs to spend multiple costs to purchase those utility tools, but also requires having space to store those utility tools. It would be appreciated to have a pool cleaning assembly kit that can be assembled to function as a pool floor scrub brush, a pool wall, step and tile brush and a pool corner brush selectively.

SUMMARY OF THE PRESENT INVENTION

[0004] The invention is advantageous in that it provides a pool cleaning assembly kit which can be assembled to form a pool floor scrub brush, a pool wall, step and tile brush and a pool corner brush selectively and adapted to be connected with various types of handle bar with different lengths.

[0005] Another advantageous of the invention is to provide a pool cleaning assembly kit which can be configured to detachably connect with a replaceable handle and two or more replaceable brush heads for various cleaning purposes to function as a pool brush.

[0006] Another advantage of the invention is to provide a pool cleaning assembly kit which is able to be configured as a pool brush assembly which main brush body also function as a docking holder to carry at least two independent brushes which can be selectively detached to work independently and to assembled with the main brush body to work collaboratively to perform various specific cleaning functions.

[0007] Another advantage of the invention is to provide a pool cleaning assembly kit which is able to be configured as a pool brush assembly configured to allow the user to replace end brush head adapted for cleaning different areas of the pool, especially such as corners, gaps, steps, clearances, uneven surfaces, and areas with irregular shapes.

[0008] Another advantage of the invention is to provide a pool cleaning assembly kit which is able to be configured as a pool brush assembly configured to allow the user to detach the end brush head to work independently to clean corners, gaps, steps, clearances, uneven surfaces, and etc. as desired with a simple push and pull action.

[0009] Another advantage of the invention is to provide a pool cleaning assembly kit which is able to be configured as a pool brush assembly configured to allow the user to replace different brush head to connect to the each end of the main brush body of the pool brush assembly.

[0010] Another advantage of the invention is to provide a pool cleaning assembly kit which is able to be configured as a pool brush assembly configured to have two different types of brush heads at two ends of the main brush body as desired for different cleaning environments and/or purposes.

[0011] Another advantage of the invention is to provide a pool cleaning assembly kit, wherein the brush elements of the two brush heads can be made of different material with respect to the brush

elements of the elongated brush frame so as to provide different brush effects to different areas of the pool, such as the flat surface and the uneven surfaces of the areas such as corner, gap, step, and clearance with irregular shapes.

[0012] Additional advantages and features of the invention will become apparent from the description which follows, and may be realized by means of the instrumentalities and combinations particular point out in the appended claims.

[0013] According to the present invention, the foregoing and other objects and advantages are attained by a pool cleaning assembly kit, comprising: [0014] a utility connector comprises a connecting dock and two end connectors integrally provided at two ends of the connecting dock respectively; [0015] a handle base comprising a handle frame detachably connected to the utility connector and configured to be detachably connected with a handle; and [0016] one or more utility tool configured to be detachably connected to at least one of the end connectors so as to integrally assemble with the utility connector to form a cleaning tool.

[0017] In one embodiment, the pool cleaning assembly kit is assembled to function as a pool brush assembly with replaceable handle and brush head arrangement that can accommodate the walls, corners and tile types of various kinds and sizes of the pool and to detach the round end brush heads from the brush body to be used independently with a handle as bowl brush or cylinder brush for smaller, curved and tubular surface.

[0018] In one embodiment, the handle base comprises: [0019] a main brush body comprising an elongated brush frame having a first end and a second end and a plurality of brush elements outwardly extended from a brushing side thereof; and [0020] a replaceable handle and brush head arrangement, which comprises: [0021] a connecting dock provided between the first end and the second end; [0022] a handle frame detachably connected to the connecting dock for detachably connecting with a handle bar; [0023] at least one end connector configured at one of the first end and the second end of the main brush body; and [0024] at least one brush head configured to be detachably connected to the at least one end connector, wherein the at one brush head comprises a connector joint and a plurality of end brush elements outwardly extended therefrom.

[0025] In one embodiment, the connecting dock comprises an elongated housing, having a receiving chamber, forming a middle portion of the main brush body, and two or more connectors formed in the receiving chamber. The handle frame having an upper end configured for connecting with the handlebar and a lower end configured to be connected with the two or more connectors of the housing of the connecting dock.

[0026] In one embodiment, the at least one end connector comprises a lock member configured to be detachably connected with the connector joint of the at least one brush head such that the end brush elements of the at least one brush head forms an end brush body of the main brush body.

[0027] In one embodiment, two end connectors are configured at the first end and the second end of the main brush body respectively and two brush heads are provided to detachably connect with lock members of the two end connectors respectively, such that the end brush elements of the two brush heads form two end brush bodies of the main brush body respectively.

[0028] In one embodiment, the pool brush assembly further comprises at least one brush head handle configured to be detachably connected to the connector joint of the at least one brush head detached from the main brush body to form an independent pool brush.

[0029] In accordance with another aspect of the invention, the present invention comprises a method of enlarging brushing area of a pool brush, comprising steps of: [0030] connecting a pair of end brush heads to two ends of a main brush body; and [0031] combining end brush elements of the two end brush heads with the brush elements of the main brush body to form two end brush bodies of the main brush body.

[0032] Still further objects and advantages will become apparent from a consideration of the ensuing description and drawings.

[0033] These and other objectives, features, and advantages of the present invention will become

apparent from the following detailed description, the accompanying drawings, and the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0034] FIG. **1** is a perspective view of a conventional pool floor scrub brush.

[0035] FIG. **2** is a perspective view of a conventional pool wall, step and tile brush.

[0036] FIG. **3** is a perspective view of a conventional pool corner brush.

[0037] FIG. **4** is a perspective view of a pool cleaning assembly kit functioning as another pool brush assembly according to a preferred embodiment of the present invention.

[0038] FIG. **5** is a perspective view of a main brush body of the pool brush assembly according to the above preferred embodiment of the present invention.

[0039] FIG. **6** is a bottom perspective view of the main brush body of the pool brush assembly according to the above preferred embodiment of the present invention.

[0040] FIG. **7** is a bottom view of the main brush body connected with two brush heads of the pool brush assembly according to the above preferred embodiment of the present invention.

[0041] FIG. **8** is perspective view of a brush head of the pool brush assembly according to the above preferred embodiment of the present invention.

[0042] FIG. **9** is another perspective view of the brush head of the pool brush assembly according to the above preferred embodiment of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0043] The following description is disclosed to enable any person skilled in the art to make and use the present invention. Preferred embodiments are provided in the following description only as examples and modifications will be apparent to those skilled in the art. The general principles defined in the following description would be applied to other embodiments, alternatives, modifications, equivalents, and applications without departing from the spirit and scope of the present invention.

[0044] Please referring to FIG. **4** to FIG. **9**, the present invention provides a pool cleaning assembly kit, which comprises: [0045] a utility connector **200** comprising a connecting dock **21** and two end connectors **23** integrally provided at two ends of the connecting dock **21** respectively; [0046] a handle base **220** comprising a handle frame **22** detachably connected to the utility connector **200** and configured to be detachably connected with a handle **40**; and [0047] one or more utility tool **300** configured to be detachably connected to at least one of the end connectors **23** so as to integrally assemble with the utility connector **200** to form a cleaning tool.

[0048] According to a preferred embodiment of the present invention, the pool cleaning assembly kit is embodied as a type of pool brush assembly, such as a pool wall, step and tile brush, which further comprises a main brush body **10** and the utility connector **200** is embodied as a replaceable handle and brush head arrangement **20**.

[0049] The main brush body **10** comprises an elongated brush frame **11** having a first end **111** and a second end **112** and a plurality of brush elements **12** outwardly extended from a brushing side of the brush frame **11**.

[0050] The replaceable handle and brush head arrangement **20** comprises the connecting dock **21** provided between the first end **111** and the second end **112** of the brush frame **11** and a handle frame **22**. The handle frame **22** is detachably connected to the connecting dock **21** for detachably connecting with the handle which can be an elongated handle bar or configured to be detachably connected with elongated handle bars with different lengths or an extensible handle bar (not shown in the drawings). The two end connectors **23** are configured at the first end **111** and the second end **112** of the main brush body **10** respectively. The one or more utility tool **300** are embodied as one

or more brush heads **30** configured to be detachably connected to the one end connector **23**.
[0051] Each of the one or more brush heads **30** comprises a connector joint **31** and a plurality of end brush elements **32** outwardly extended therefrom. According to the second preferred embodiment, the two end connectors **23** are provided at the first end **111** and the second end **112** of the brush frame **11** of the main brush body **10** respectively and the two brush heads **30** are provided to detachably connect with the two end connectors **23** respectively, such that the end brush elements **32** of the two brush heads **30** form two end brush bodies **101**, **102** extended from the first end **111** and the second end **112** of the main brush body **10** respectively.

[0052] As shown in FIG. 5 and FIG. 6, each of the end connectors **23** comprises a lock member **231** configured to be detachably connected with the connector joint **31** of the respective brush head **30** and at least one locking hole **2312** formed on top of lock member **231**, as shown in FIG. 7, such that the end brush elements **32** of the brush head **30** forms the end brush body **101**, **102** of the main brush body **10**, as shown in FIG. 4.

[0053] According to the preferred embodiment, as shown in FIG. 7 and FIG. 8, the connector joint **31** of each of the brush head **30** is a tubular body having a receiving socket **311** and comprises at least one locking button **312** provided on the connector joint **31** and extended into the receiving socket **311**, as shown in FIG. 8, wherein each of the receiving socket **311** is a rectangular socket having a U-shape bottom opening slot **3111** rendering the connector joint **31** a resilient body to fittingly receive and grasp the respective lock member **231** therein.

[0054] Correspondingly, referring to FIG. 6 to FIG. 7, according to the preferred embodiment, the lock member **231** of each of the end connectors **23** comprises a joint arm **2311** extended from the respective first or second end **111**, **112** of the main body **11**, wherein each of the two joint arms **2311** has a rectangular shape and size configured to be fittingly inserted into the receiving socket **311** of the respective connector joint **31** of the brush head **30**. To connect the brush head **30** (utility pool **300**) to the main brush body **20** (utility connector **200**), the joint arm **2311** of the lock member **231** of the end connector **23** is fittingly inserted into the receiving socket **311** until the locking button **312** is inserted into the locking hole **2312** such that the brush head **30** is locked and connected with the end connector **23** of the main brush body **20**. To detach the brush head **30** from the main brush body **20**, simply detach the locking button **312'** from the receiving socket **2312** to release the locking of the lock member **231** and the end connector **23**. Then, the brush head **30** is able to be pulled out from the lock member **231** and detached from the main brush body **20**.

[0055] In one embodiment, the at least one locking button **312**, which can be a resilient locking button, comprises a button head **313** and is configured to be clicked and locked in the locking hole **2312** until the resilient locking button **312** is inserted into the locking hole **2312** so as to lock and connect the brush head **20** with the end connector **23** of the main brush body **20**. To detach the brush head **30** from the main brush body **20**, simply pull up the button head **313** to move the resilient locking button **312** from the locking hole **2312** to release the locking of the lock member **231** and the end connector **23**. Then, the brush head **30** is able to be pulled out from the lock member **231** and detached from the main brush body **20**.

[0056] In one embodiment, the connector joint **31** of each of the brush head **30** has a button hole **310**, as shown in FIG. 8 and FIG. 9, for the locking button **312** extending from the button head **313** into the receiving socket **311**. The button hole **310** can be embodied as a threaded hole and the locking button **312** is embodied as a threaded rod extended from the button head **313** and screwed through the button hole **31** into the receiving socket **311**, such that to connect the brush head **30** (utility pool **300**) to the main brush body **20** (utility connector **200**), fittingly insert the joint arm **2311** of the lock member **231** of the end connector **23** into the receiving socket **311**, rotate locking button **312** to screw down the locking button **312** to insert into the locking hole **2312** such that the brush head **30** is locked and connected with the end connector **23** of the main brush body **20**. To detach the brush head **30** from the main brush body **20**, simply rotate the button head **313** in an opposite direction to unscrew the locking button **312** from the receiving socket **2312** to release the

locking of the lock member **231** and the end connector **23**. Then, the brush head **30** is able to be pulled out from the lock member **231** and detached from the replaceable handle and brush head arrangement **20**.

[0057] It is appreciated that a plurality of connector brush elements **121** is outwardly extended from a bottom side of the lock member **231** of each of the end connector **23** such that the connector brush elements **121** of the two lock member members **231** substantially increase a total length of the main brush body **10**. In addition, the connector brush elements **121** can be extended out through the bottom opening slot **3111** of the receiving socket **311** of the connector joint **31** and extended between the plurality of brush elements **12** of the main brush body **10** and the end brush elements **32** of the brush head **30** when the two brush heads **30** are connected to the two end connectors **23** respectively.

[0058] Referring to FIG. 5 to FIG. 7, in order to ensure a secure and rigid connection between the main brush body **20** and the brush head **30** through the corresponding end connector **23** and the connector joint **31**, it is preferred that the width of the connecting dock **21** of the main brush body **20** is equal to the width of the connector joint **31**, wherein the brushing force applied to the main brush body **20** can be securely applied to the left and right brush heads **30** respectively connected to the two lock members **231** of the two connectors **23** formed at the two left and right ends **111**, **112** of the brush frame **11** of the main brush body **20**. According to the preferred embodiment, the width of each of the joint arms **2311** of the two end connectors **23** is smaller than the width of the connecting dock **21**, such that after the joint arms **2311** are fittingly inserted into the receiving sockets **311** of the joint portions **311** of the connector joints **31** and locked by the locking buttons **312** respectively, the main brush body **20** and the brush heads **30** are integrally connected to form the pool cleaning assembly with the same width.

[0059] It is appreciated that the size of the joint arm **2311** is made fitting with the size of the receiving socket **311** of the lock member **231**, such that when the joint arm **2311** is fittingly inserted into the receiving socket **311**, the joint arm **2311** and the lock member **231** form an interconnecting configuration that not only rigidly connected the brush head **30** to the replaceable handle and brush head arrangement **20**, but also the walls of the joint arm **2311** and the lock member **231** substantially thicken and strengthen the first or second end **111**, **112** of the brush frame **11** while securely connecting the two brush heads **30** with replaceable handle and brush head arrangement **20** to form the pool brush assembly.

[0060] Referring to FIG. 6 and FIG. 7, two or more connection holes **211** are spacedly formed in the connecting dock **21**, such that the handle frame **22** can be detachably mounted to the connecting dock **21** by means of screwing, buckling, or other connecting means.

[0061] Referring to FIG. 8 and FIG. 9, each of the brush head **30** comprises head frame **33** and the end brush elements **32** are downwardly extended from a bottom of the head frame **33**. The head frame **33** can be a straight body such that the brush heads **30** can be used to further lengthen the main brush body **10**. According to the second preferred embodiment, the head frame **33** comprises a joint portion **331** to configure the connector joint **31** and a curved portion **332** and the end brush elements **32** are at least downwardly extended from the curved portion **332** to form a curved brush body for brushing corners area.

[0062] It should be understood that the joint portion **331** can be made to have an elongated shape so that additional end brush elements **32** can also be extended downwardly from the joint portion **331** to lengthen the main brush body **10** for brushing larger area of the pool walls and the pool bottom.

[0063] Like the above first preferred embodiment, the detached brush head **30** as shown in FIG. 8 can be worked as an independent curved pool corner brush, wherein the pool brush assembly according to the second preferred embodiment of the present invention further comprises at least one brush head handle **50** configured to be detachably connected to the connector joint **31** of the brush head **30**. The brush head handle **50** is an elongated handle bar having an end configured like

the lock member **231** of the end connector **23** for detachably connecting with the connector joint **31**'.

[0064] One skilled in the art will understand that the embodiment of the present invention as shown in the drawings and described above is exemplary only and not intended to be limiting.

[0065] It will thus be seen that the objects of the present invention have been fully and effectively accomplished. The embodiments have been shown and described for the purposes of illustrating the functional and structural principles of the present invention and is subject to change without departure from such principles. Therefore, this invention includes all modifications encompassed within the spirit and scope of the following claims.

Claims

1. A pool cleaning assembly, which comprises: a utility connector comprising a connecting dock and two end connectors integrally provided at two ends of the connecting dock respectively, wherein the connecting dock comprises a main brush body comprising an elongated brush frame having a first end and a second end and a plurality of brush elements outwardly extended from a brushing side thereof, and a replaceable arrangement which comprises at least one end connector configured at one of the first end and the second end of the brush frame of the main brush body; a handle base comprising a handle frame detachably connected to the utility connector and configured to be detachably connected with a handle; and one or more utility tools configured to be detachably connected to at least one of the end connectors so as to integrally assemble with the utility connector to form a cleaning tool, wherein the one or more utility tools is a brush head configured to be detachably connected to the end connector such that a width of each of the one or more brush heads is generally equal to a width of the elongated brush frame of the main brush body, wherein each of the one or more brush heads comprises a connector joint and a plurality of end brush elements outwardly extended therefrom.

2. The pool cleaning assembly, as recited in claim 1, wherein the replaceable arrangement further comprises a connecting dock provided between the first end and the second end of the brush frame and a handle frame detachably connected to the connecting dock for detachably connecting with the handle bar.

3. The pool cleaning assembly, as recited in claim 2, wherein each of the end connectors comprises a lock member configured to be detachably connected with the connector joint of the respective brush head and at least one locking hole formed on top of lock member, such that the end brush elements of the brush head form the end brush body of the main brush body.

4. The pool cleaning assembly, as recited in claim 3, wherein the connector joint of each of the brush head is a tubular body having a receiving socket and comprises at least one locking button provided on the connector joint and extended into the receiving socket, wherein each of the receiving socket is a socket having a U-shape bottom opening slot rendering the connector joint a resilient body to fittingly receive and grasp the respective lock member therein.

5. The pool cleaning assembly, as recited in claim 4, wherein a width of each of the joint arms of the two end connectors is smaller than a width of the connecting dock, such that after each of the joint arms is fittingly inserted into the receiving socket of the joint portion of the corresponding connector joint, the width of the connecting dock of the main brush body is equal to the width of the connector joint.

6. The pool cleaning assembly, as recited in claim 4, wherein the lock member of each of the end connectors comprises a joint arm extended from the respective first or second end of the main body, wherein each of the two joint arms has a shape and size configured to be fittingly inserted into the receiving socket of the respective connector joint of the brush head, such that to connect the brush head to the main brush body, the joint arm of the lock member of the end connector is inserted into the receiving socket until the locking button is inserted into the locking hole such that

the brush head is locked and connected with the end connector of the main brush body, and that to detach the brush head from the main brush body, the locking button is detached from the receiving socket to release the locking of the lock member and the end connector, so that the brush head is able to be pulled out from the lock member and detached from the main brush body.

7. The pool cleaning assembly, as recited in claim 5, wherein the lock member of each of the end connectors comprises a joint arm extended from the respective first or second end of the main body, wherein each of the two joint arms has a shape and size configured to be fittingly inserted into the receiving socket of the respective connector joint of the brush head, such that to connect the brush head to the main brush body, the joint arm of the lock member of the end connector is inserted into the receiving socket until the locking button is inserted into the locking hole such that the brush head is locked and connected with the end connector of the main brush body, and that to detach the brush head from the main brush body, the locking button is detached from the receiving socket to release the locking of the lock member and the end connector, so that the brush head is able to be pulled out from the lock member and detached from the main brush body.

8. The pool cleaning assembly, as recited in claim 4, wherein the at least one locking button, which is a resilient locking button, comprises a button head and is configured to be clicked and locked in the locking hole until the resilient locking button is inserted into the locking hole so as to lock and connect the brush head with the end connector of the main brush body, such that to detach the brush head from the main brush body, the button head is pulled up to move the resilient locking button from the locking hole to release the locking of the lock member and the end connector, so that the brush head is able to be pulled out from the lock member and detached from the main brush body.

9. The pool cleaning assembly, as recited in claim 5, wherein the at least one locking button, which is a resilient locking button, comprises a button head and is configured to be clicked and locked in the locking hole until the resilient locking button is inserted into the locking hole so as to lock and connect the brush head with the end connector of the main brush body, such that to detach the brush head from the main brush body, the button head is pulled up to move the resilient locking button from the locking hole to release the locking of the lock member and the end connector, so that the brush head is able to be pulled out from the lock member and detached from the main brush body.

10. The pool cleaning assembly, as recited in claim 4, wherein the connector joint of each of the brush head has a button hole for the locking button extending from the button head into the receiving socket, wherein the button hole is a threaded hole and the locking button is a threaded rod extended from the button head and screwed through the button hole into the receiving socket, such that to connect the brush head to the main brush body, the joint arm of the lock member of the end connector is inserted into the receiving socket, locking button is rotated to screw down the locking button to insert into the locking hole such that the brush head is locked and connected with the end connector of the main brush body, such that to detach the brush head from the main brush body, the button head is rotated in an opposite direction to unscrew the locking button from the receiving socket to release the locking of the lock member and the end connector, so that the brush head is able to be pulled out from the lock member and detached from the replaceable handle and brush head arrangement.

11. The pool cleaning assembly, as recited in claim 5, wherein the connector joint of each of the brush head has a button hole for the locking button extending from the button head into the receiving socket, wherein the button hole is a threaded hole and the locking button is a threaded rod extended from the button head and screwed through the button hole into the receiving socket, such that to connect the brush head to the main brush body, the joint arm of the lock member of the end connector is inserted into the receiving socket, locking button is rotated to screw down the locking button to insert into the locking hole such that the brush head is locked and connected with the end connector of the main brush body, such that to detach the brush head from the main brush body, the

button head is rotated in an opposite direction to unscrew the locking button from the receiving socket to release the locking of the lock member and the end connector, so that the brush head is able to be pulled out from the lock member and detached from the replaceable handle and brush head arrangement.

12. The pool cleaning assembly, as recited in claim 6, wherein a plurality of connector brush elements is outwardly extended from a bottom side of the lock member of each of the end connector such that the connector brush elements of the two lock member members increase a total length of the main brush body.

13. The pool cleaning assembly, as recited in claim 7, wherein a plurality of connector brush elements is outwardly extended from a bottom side of the lock member of each of the end connector such that the connector brush elements of the two lock member members increase a total length of the main brush body.

14. The pool cleaning assembly, as recited in claim 10, wherein the connector brush elements are extended out through the bottom opening slot of the receiving socket of the connector joint and extended between the plurality of brush elements of the main brush body and the end brush elements of the brush head when the two brush heads are connected to the two end connectors respectively.

15. The pool cleaning assembly, as recited in claim 11, wherein the connector brush elements are extended out through the bottom opening slot of the receiving socket of the connector joint and extended between the plurality of brush elements of the main brush body and the end brush elements of the brush head when the two brush heads are connected to the two end connectors respectively.

16. The pool cleaning assembly, as recited in claim 8, wherein a plurality of connector brush elements is outwardly extended from a bottom side of the lock member of each of the end connector such that the connector brush elements of the two lock member members increase a total length of the main brush body.

17. The pool cleaning assembly, as recited in claim 9, wherein a plurality of connector brush elements is outwardly extended from a bottom side of the lock member of each of the end connector such that the connector brush elements of the two lock member members increase a total length of the main brush body.

18. The pool cleaning assembly, as recited in claim 16, wherein the connector brush elements are extended out through the bottom opening slot of the receiving socket of the connector joint and extended between the plurality of brush elements of the main brush body and the end brush elements of the brush head when the two brush heads are connected to the two end connectors respectively.

19. The pool cleaning assembly, as recited in claim 17, wherein the connector brush elements are extended out through the bottom opening slot of the receiving socket of the connector joint and extended between the plurality of brush elements of the main brush body and the end brush elements of the brush head when the two brush heads are connected to the two end connectors respectively.
